Experiment-18

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Course Code: -MLA0305 **Slot-**B

Title: Implementation of a Multi-Agent Reinforcement Learning System

Aim:
To develop a Multi-Agent Reinforcement Learning system in which multiple agents cooperate or compete to maximize cumulative rewards.

Procedure:

1. Define a multi-agent environment.
2. Initialize multiple independent agents.
3. Define the state and action space for each agent.
4. Assign individual or shared rewards.
5. Allow agents to interact with the environment simultaneously.
6. Train agents using suitable RL algorithms.
7. Record individual and total rewards.
8. Analyze cooperative or competitive behavior.
9. Evaluate the final policies.
10. Compare the performance of multiple agents.

```python
import random
score=[0,0]
for ep in range(10):
 a=[random.choice([0,1]),random.choice([0,1])]; r=int(a[0]==a[1]); score[0]+=r;score[1]+=r;print(ep+1,a,r)
print('Shared rewards:',score)
```

```
1 [1, 0] 0
2 [0, 1] 0
3 [0, 0] 1
4 [0, 1] 0
5 [0, 0] 1
6 [0, 0] 1
7 [1, 1] 1
8 [0, 0] 1
9 [0, 1] 0
10 [0, 1] 0
Shared rewards: [5, 5]
```

Result:

A multi-agent Reinforcement Learning system was successfully developed. The agents learned cooperative or competitive strategies based on their reward structure.